

SUHAS MORISETTY

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EDUCATION

Columbia University

Master of Science in Computer Science

New York, NY

Expected Dec 2026

Indian Institute of Technology Madras (IITM)

Dual Degree (Electrical Engineering & Data Science)

Chennai, IN

June 2022

PROFESSIONAL EXPERIENCE

Capital One

AI Engineering Intern

New York, NY

June 2026 – Aug 2026

- Designed a workload-aware KV cache eviction policy that at least matches LRU and outperforms it where LRU falls short.
- Pinpointed LRU's failure modes across realistic workloads, motivating an adaptive policy.
- Built a shadow trace replayer enabling safe, online policy selection with zero impact on live traffic.
- Implemented and benchmarked multiple eviction policies (LFU, S3-FIFO, priority-based) in vLLM; drove the work end-to-end, with the full effort in preparation for publication.

Microsoft

Data and Applied Scientist

Bangalore, IN

July 2022 – May 2024

- Developed and deployed a multitask non-autoregressive generative retrieval model, improving keyword retrieval recall by 10% offline and delivering a 1% revenue lift for Bing sponsored search advertisements.
- Unified dense and generative retrieval models with a shared encoder, roughly doubling ROI per GPU compute.
- Designed and implemented a novel keyword extraction pipeline from search logs and ad clicks for Bing's keyword-less dynamic search ads, resulting in a 20% revenue increase with no quality degradation.
- Led brand-sensitive keyword filtering efforts for int'l markets, reducing brand mismatch defects by 20% through online A/B test experiments on test traffic.
- Overall, delivered multimillion dollar annual revenue for Bing Ads int'l markets, validated via online experiments.

Adobe

Applied Scientist

Bangalore, IN

May 2024 – July 2025

- Implemented training-free adaptation techniques (e.g., inversion, SDEdit) on Adobe Firefly foundational models to convert photographic raster images into vector-style graphics.
- Created a vector graphic aesthetic score predictor, translating abstract artist and designer feedback into actionable rules, generating unbiased labeled data, and modeling a predictor using preference data.
- Fine-tuned vision-language models to build a training data-filtering pipeline, leading to significant qualitative improvements to Adobe Illustrator's Generative Expand feature.
- Identified failure modes in the Turntable (image-to-3D) model and trained a classifier to flag unsuitable cases, achieving greater than 90% accuracy, and applied knowledge distillation to create lightweight models for on-device deployment.

RESEARCH EXPERIENCE

Microsoft: Query Rewriting Benchmark

Aug 2023 – Dec 2023

- Constructed a checklist-based evaluation to identify failure modes in short text query keyword retrieval models.
- Utilized GPT-4 with self-instruct prompting to iteratively generate candidate test cases for each checklist criterion.
- Evaluated a range of generative and dense retrieval models, surfacing roughly 10 actionable insights for improving retrievers.

IITM: Restoration of Dark Stereo Images (Dr. Kaushik Mitra)

Sep 2021 – June 2022

- Devised a real-time U-Net-based network for enhancing stereo low-light images.
- Proposed a novel stereo consistency loss to ensure stereo constraints remain intact in enhanced images.
- One of the first works addressing stereo extreme low-light enhancement; led to a publication in WACV '23.
- Ideated a new dataset of paired well-lit and low-lit images and collected nighttime data through custom electrical and software solutions.

RELEVANT SKILLS

C++, Python, SQL, Matlab, PyTorch, OpenCV, ONNX, CoreML, Hugging Face, NLP, Information Retrieval, PEFT, Large Language Models, Diffusion Models.